

Software Engineer with 6+ years of experience facilitating engineering solutions with a wide range of software applications. Having a strong communication, leadership and technical background. Looking forward to leverage my skills and share knowledge with fellow software enthusiasts.

Education

2012–2016 **Sacs MAVMM Engineering College,**
Computer science engineering, Bachelor of Engineering.

Experience

- June 2021– **Staff software engineer,** IGNITARIUM TECHNOLOGY SOLUTIONS.
Ongoing
- Implemented model evaluation metrics for mobilenetv1, ssdmobilenetv1 and other custom developed models.
 - Client interaction to gather requirements and present ideas.
 - Implemented workflow automation (CI/CD) with Jenkins and Azure services.
 - Created unit test cases with pytest module.
 - Developed software application and deployed using docker and kubernetes.
 - Developed Azure functions to monitor kubernetes pod's health.
 - Worked on Azure logic apps and Azure service bus to trigger emails to customers based on specific event.
 - Developed modules for Azure storage container to read/write blobs.
 - Worked on IOT devices and developed applications.
- Sep 2016–June 2021 **Lead Machine learning Engineer,** OPTISOL BUSINESS SOLUTIONS.
Managed AI team, developed software applications and built deep learning models.
- Developed AI models for image classification, object detection and pose estimation with optimal accuracy.
 - Deployed software applications with cloud services.
 - Developed AWS lambda triggers to pre-process image files in S3 bucket.
 - Written deployment scripts using Docker.
 - Developed RESTful APIs and deployed using AWS Elastic registries and containers.
 - Estimated software requirements and resource availability with optimal timeline.
 - Hired resources and managed a team of 5 developers.

Projects

- July 2021– **Cloud based model evaluation tool,** Sony India software solutions, Bangalore.
Ongoing
- Developed **micro services** using **Python** and **Azure services**.
 - Developed **Restful APIs** using **Flask** and **FastAPI** python web frameworks and **MySQL** sql as a database.
 - Collected images from cloud and processed with dynamic image processing logic using **OpenCV** and **Pillow** libraries.
 - Implemented automation for an internal software module and integrated as a cloud service using python and deployed using **Kubernetes**.
 - Written model evaluation script for PYCOCO and PASCAL VOC dataset to evaluate Object detection models.
 - Implemented a generic module to parse Object detection and image classification models.
 - Handled **CI/CD** using **Jenkins** with best practices.
 - Implemented unit test automation and code coverage with **Pytest** and integrated the same CI/CD deployment.
 - Followed **pep8** Standards.
 - Developed solution to abide the standards followed at Sony India software solutions.
 - Supported the team to build a **micro services architecture** using Microsoft Azure.
 - Collected requirements directly from client and estimated timeline for multiple releases.
- Dec 2021– **Offline model evaluation tool,** Sony India software solutions, Bangalore.
Ongoing
- Developed a **Python** application with **IOT** using paramiko library.
 - Provided R&D support on **Python UI frameworks** and developed UI using **PyQT** library.
 - Developed solution to abide the standards followed at Sony India software solutions.
 - Implemented **YAML parsing** to process the inputs from user.
 - Implemented multiprocessing technique to utilise multiple IOT devices and process in a small amount of time.
 - Managed a resource and mentored on development ideas.
 - Collected requirements directly from client and estimated timeline for multiple releases.

- Feb 2019 - **Gesture recognition for biblical phrases**, *Optisol business solutions, Chennai.*
- June 2021
- Implemented a combination of **CNN + LSTM** to identify human poses and gestures on a real-time.
 - Taken responsibility of collecting dataset, training new models and inferring them on real-time.
 - Developed back-end using **python FastAPI web framework**.
 - Achieved 90% accuracy on identifying the gestures in the live inference.
 - Proposed a system using **React JS, Fast API, Web Sockets, MySQL using micro-services architecture**.
- Dec 2018 - **Facial recognition system**, *Optisol business solutions, Chennai.*
- Mar 2020
- Implemented this internal project for Optisol business solutions using **Python & flask framework**.
 - Collected 200 employee images and trained a VGG-net.
 - Explored **AWS sagemaker** for training and deployment of the model.
 - Developed REST Api to feed new images and train the with the existing dataset (ML Ops).
 - Used **S3 bucket** to store and retrieve the files.
 - Developed scripts to pre-process images using **AWS lambda**.
 - Connected MySQL database to the system to store the data.
 - Architect the system using **AWS micro-services architecture** and **Docker containers**.
 - Internally used this system to record in / out time of the employees.
- May 2018 - **Person detector on a forklift vehicle**, *Optisol business solutions, Chennai.*
- Dec 2020
- Developed a computer vision based **Python** application.
 - Sanity check API developed using **Flask** framework.
 - Collected 1000s of human images on the site and used **MySQL** database on storing the meta data.
 - Implemented transfer learning and trained a model to identify humans with **95% accuracy** using **Tensorflow, Keras framework**.
 - Performed Model Quantization to the model using **Intel's OpenVino framework** and achieved over 6 times higher FPS than the normal execution.
 - Deployed this application on an **Intel NUC**.
 - Implemented post processing logic for human detection and calculated real-time distance from the camera.
 - Triggered run-time alarms by accessing GPIO pins using python serial library.
- Jan 2017-April 2018 **Lovely childcare**, *Optisol business solutions, Madurai.*
- Provided overall time estimation to the client.
 - Developed application back-end using **Django web framework** in **python**.
 - Used **MySQL** database sql database integrated with Django ORM.
 - Developed front-end using **ReactJS**.
 - Designed and developed the project architecture and deployed using **AWS EC2**.
 - Written APIs to securely upload files in **Amazon's S3 bucket**.
- Sep 2016-Apr 2018 **Human resource management system**, *Optisol business solutions, Madurai.*
- Provided overall time estimation to the company internal managers.
 - Developed application back-end using **Django web framework** in **python**.
 - Used **MySQL** sql database integrated with Django ORM.
 - Developed front-end using **AngularJS**.
 - Designed and developed the project architecture and deployed using **AWS EC2**.

Achievements

- June 2023 **Leveled up**, *Ignitarium technologies.*
- Promoted as a **Staff software engineer**.
- July 2022 **Sony Quarterly award**, *Sony India software centre.*
- Secured Best performance award for playing a valuable role in various development activities.
- May 2022 **Spot award for delivery excellence**, *Sony India software centre.*
- spot award for on-time delivery of the products.
- Feb 2022 **Best Team award**, *Sony India software centre.*
- Scored Best Team award among the SONY india team for excellence in product development.
 - Received appreciations from Lead and manager on being a team player.
- Oct 2018 **Quick Draw! Doodle Recognition Challenge**, *Kaggle.*
- Ranked in the TOP 29% in the Doodle Recognition Challenge.
 - Trained an ImageNet model from scratch, by constructing images from stokes.

Skills

Languages Python(A), Javascript(B), SQL(B), Groovy(B)

Frameworks Keras, PyTorch, Tensorflow, Django, FastAPI, Flask
WebD HTML/CSS, JavaScript, VueJS
Mobile Flutter (Basic self learned)
Utilities Git, VS Code, Jupyter Notebook, Jenkins
Cloud Azure cloud services(Storage containers, Container registry, Logic apps, IOT hub, Service bus, Azure functions,), AWS(EC2, Lambda, ECR, Micro-services)
Communication English(SRW), Tamil(SRW)

Relevant Courses

Convolution Neural Networks for Visual Recognition, Introduction to Machine Learning, Introduction to Statistics and Probability, Data structures in python

Extra Curricular

- Apr 2018 **Developer Meetups**, *Executive Member*.
Participated multiple Software development meetups and Presented ideas.
- Oct 2017 **Optisol internal Hackathon**, *Executive Member*.
Developed a POC application on Image classification using tensorflow.

Interests

- Badminton
- Painting
- Swimming
- Gaming